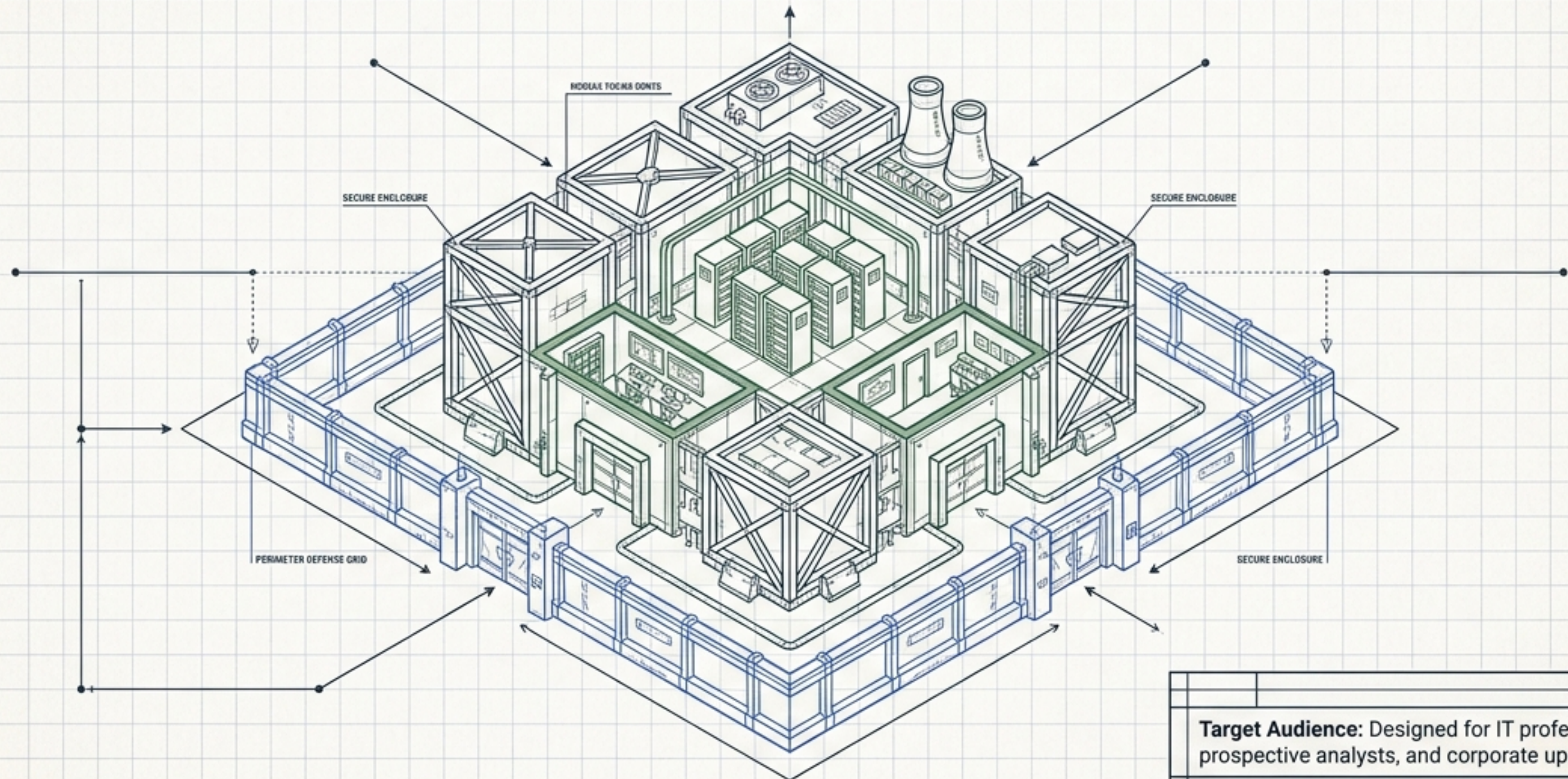


The Security Blueprint: Assets, Threats, and Vulnerabilities

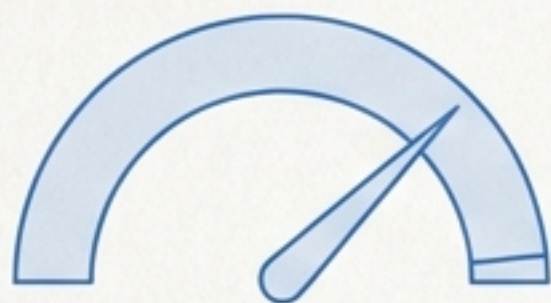
A structural breakdown of Course 5 in the Google Cybersecurity Professional Certificate.



Target Audience: Designed for IT professionals, prospective analysts, and corporate upskilling.

Operational Parameters: Course Snapshot

Scale & Trust



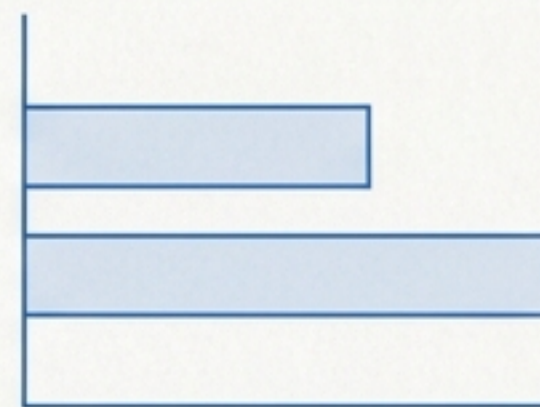
351,705+ Enrolled

4.8/5 Rating (4,112 Reviews)

98% Learner Satisfaction



Pace & Commitment



2-Week Flexible Schedule

~10 Hours per week

Accessibility



Beginner Level

No prior experience required



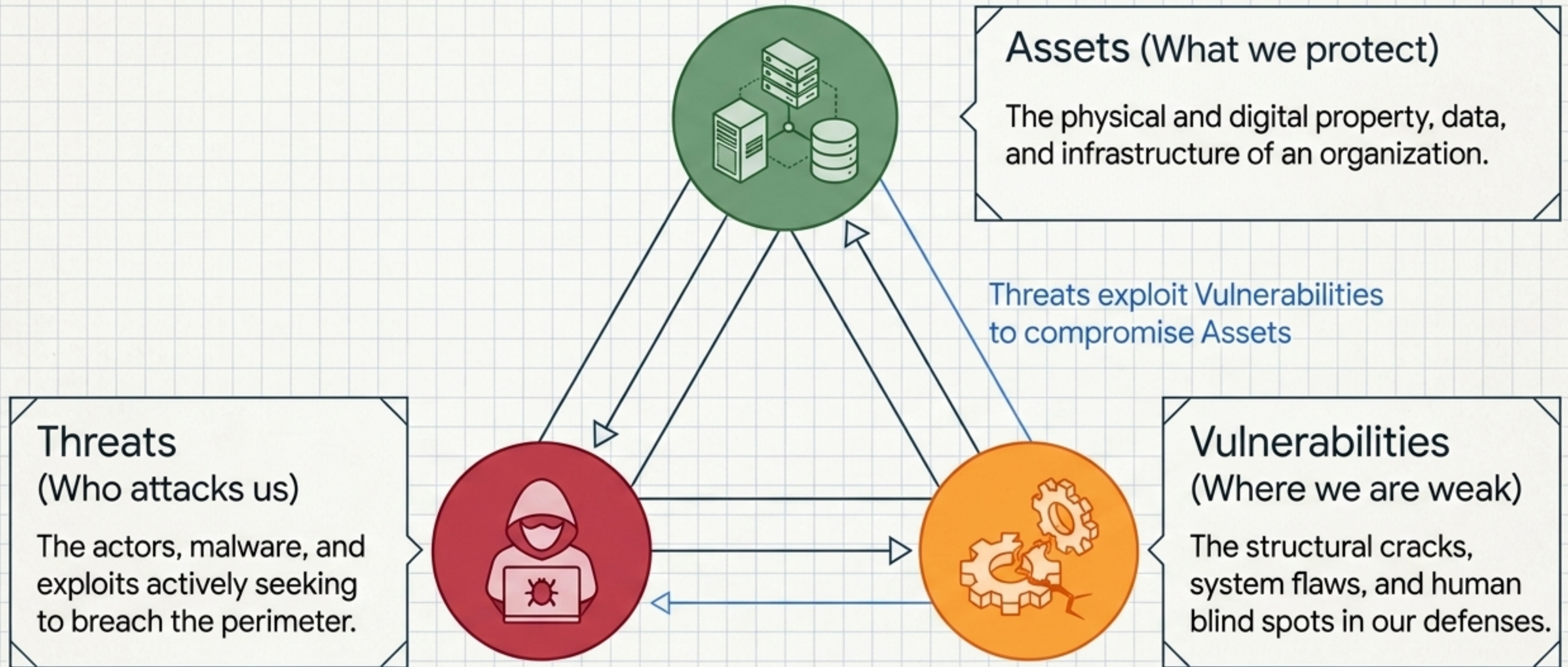
Application



26 Hands-On Assignments

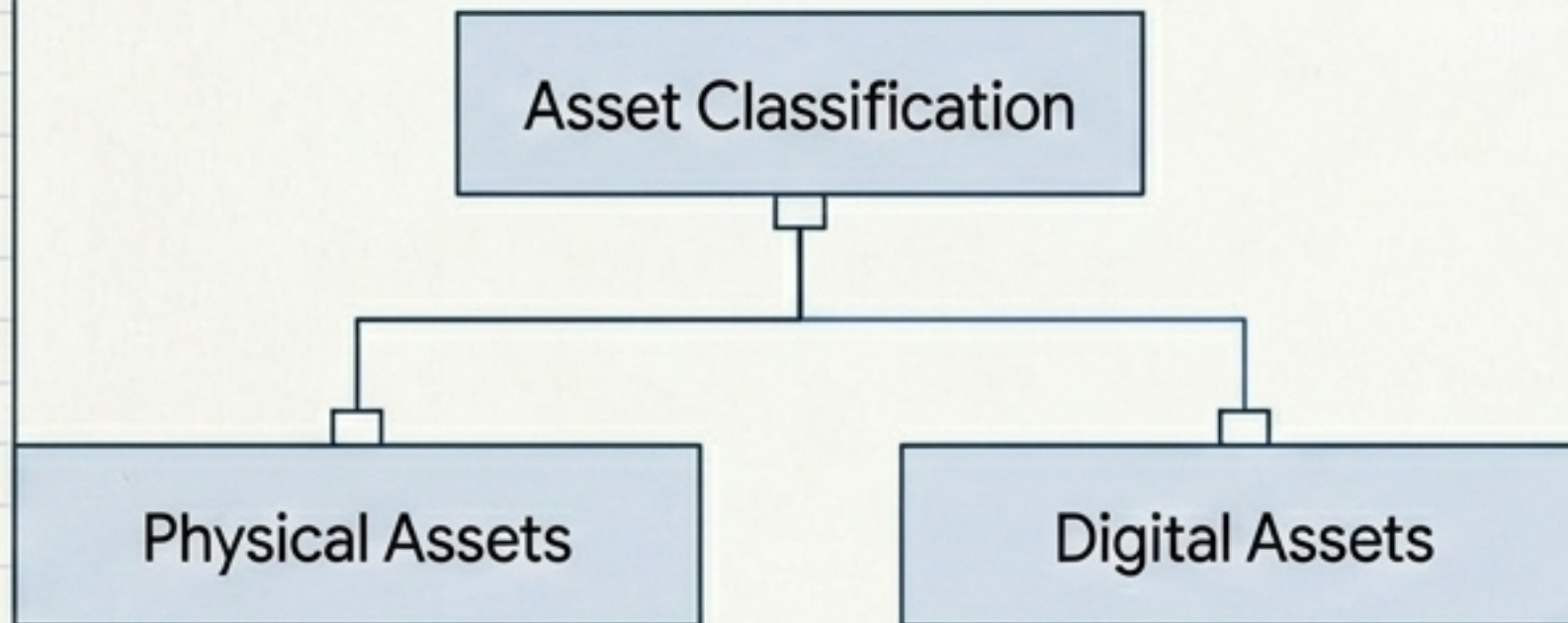
Taught by Google Career
Certificates Experts

The Cybersecurity Triad: A Continuous Dynamic



Module 01: Defining and Classifying Assets

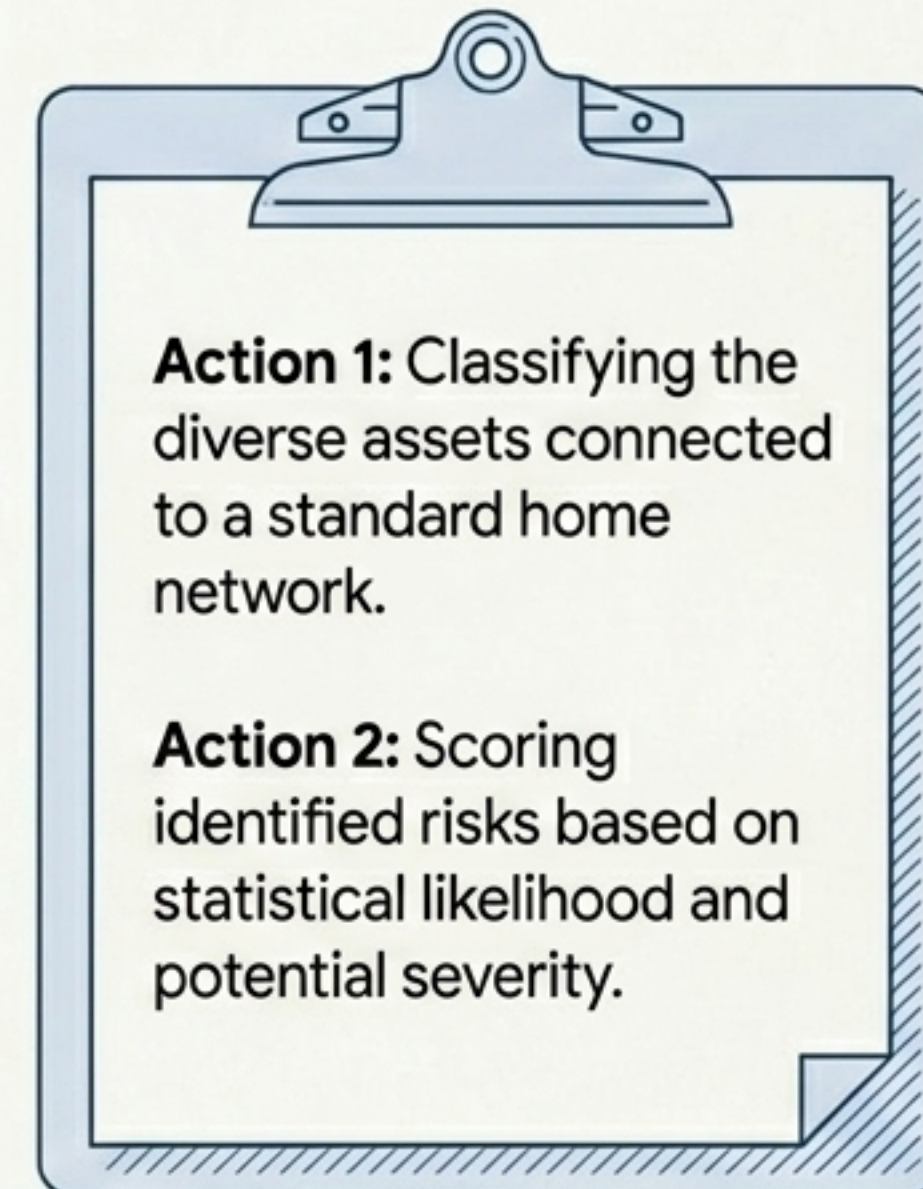
Theoretical Framework (4 Hours)



Core Concepts:

- The NIST Cybersecurity Framework (CSF) standards.
- Mapping the data lifecycle in a cloud-first world.

Practical Application



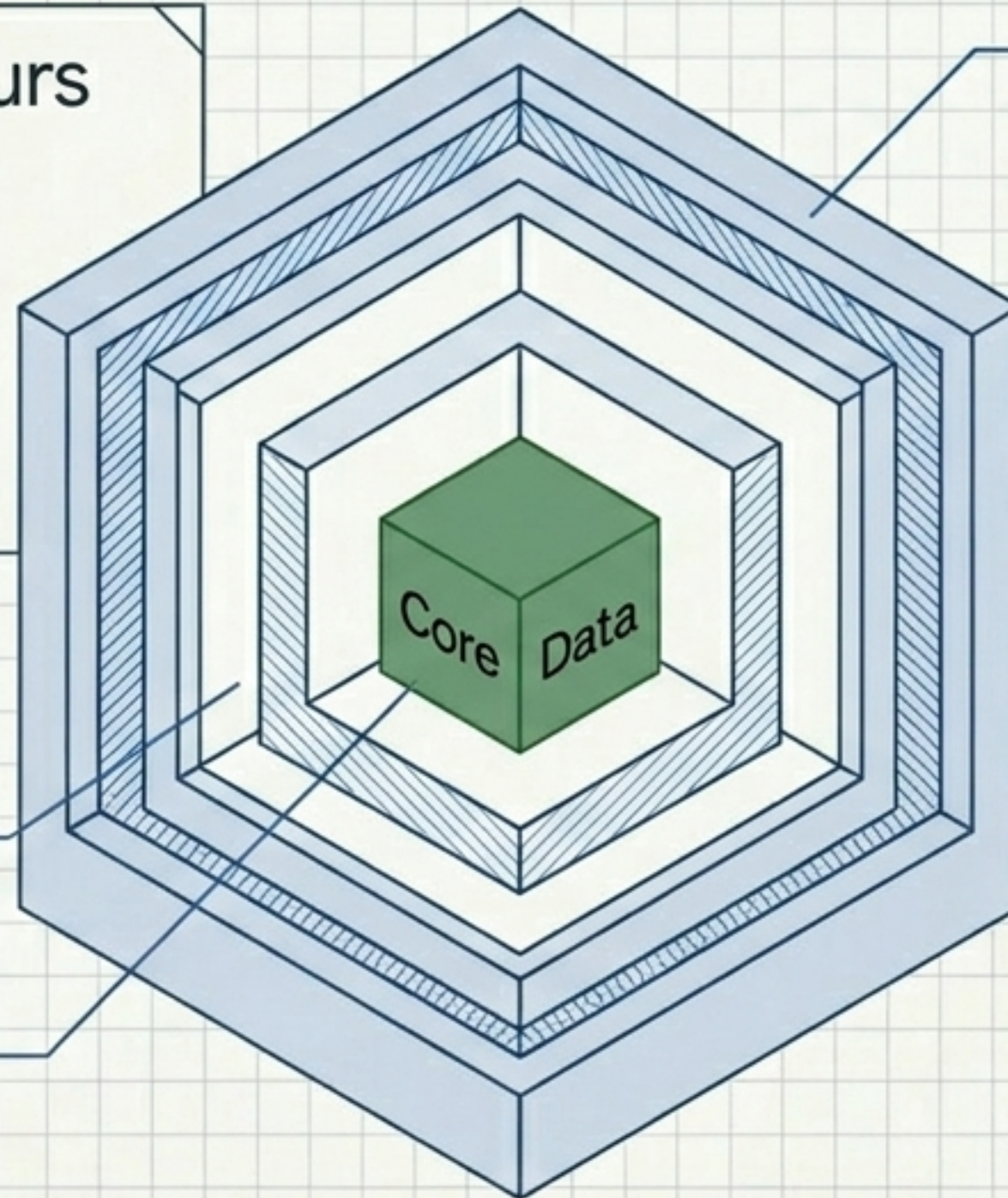
Module 02: Controls, Cryptography, and Protection

Theoretical Focus: 6 Hours

- The AAA Security Concept
- Symmetric vs. Asymmetric encryption
- Public Key Infrastructure (PKI)
- Evolution of hash functions

Authorization
(What can you do?)

Accounting
(What did you do?)



Authentication (Who are you?)

Hands-On Execution



- Decrypting an encrypted message using cryptographic tools.

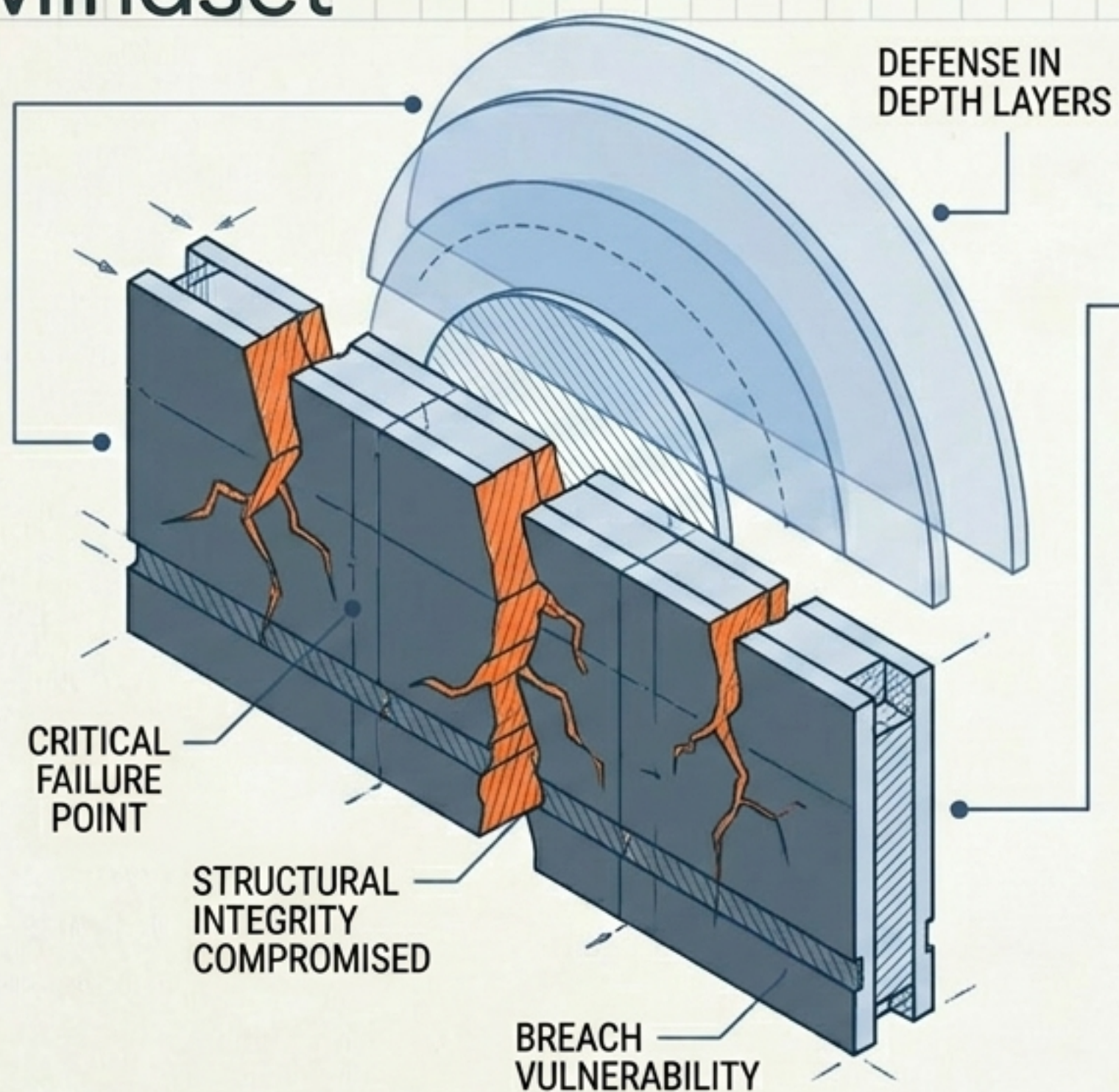


- Creating hash values to verify data integrity.



- Upgrading authentication protocols for a simulated business.

Module 03: Vulnerability Management & The Attacker Mindset



Theoretical Framework (5 Hours)

- System Flaws: Understanding the OWASP Top 10 and CVE tracking.
- Tactics: Open Source Intelligence (OSINT) and penetration testing.
- Mitigation: The critical importance of continuous updates.

Practical Application

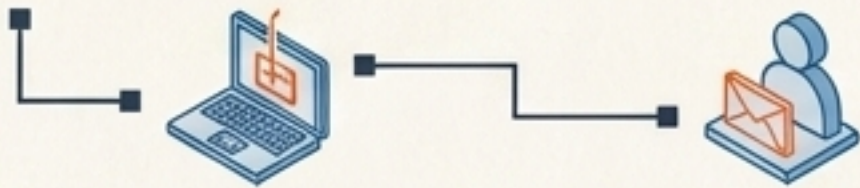
- Analyzing a vulnerable system for a small business to identify exploitable gaps.
- Identifying and mapping specific attack vectors introduced by a rogue USB drive.

Module 04: Threat Intelligence and Attack Vectors

Hands-On Execution (4 Hours)

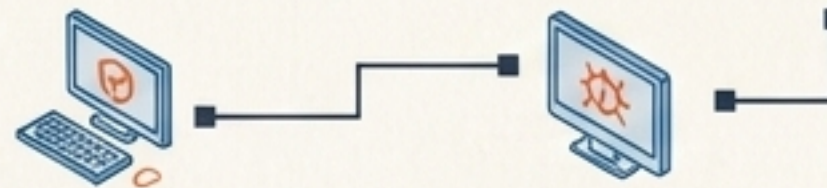
Human Vectors

Social Engineering & Phishing
(The criminal art of persuasion)



Software Vectors

Malicious Software
(Malware & Cryptojacking)



Web Vectors

Web Exploits
(Cross-site scripting / XSS)



Data Vectors

Database Gaps
(Injection attacks)



- Filtering and analyzing malicious emails.
- Applying the PASTA threat model framework to a simulated application.

Proactive Modeling: Process for Attack Simulation and Threat Analysis (PASTA)

1. Define Objectives

2. Define Technical Scope

3. Application Decomposition

4. Threat Analysis

5. Vulnerability & Weakness Analysis

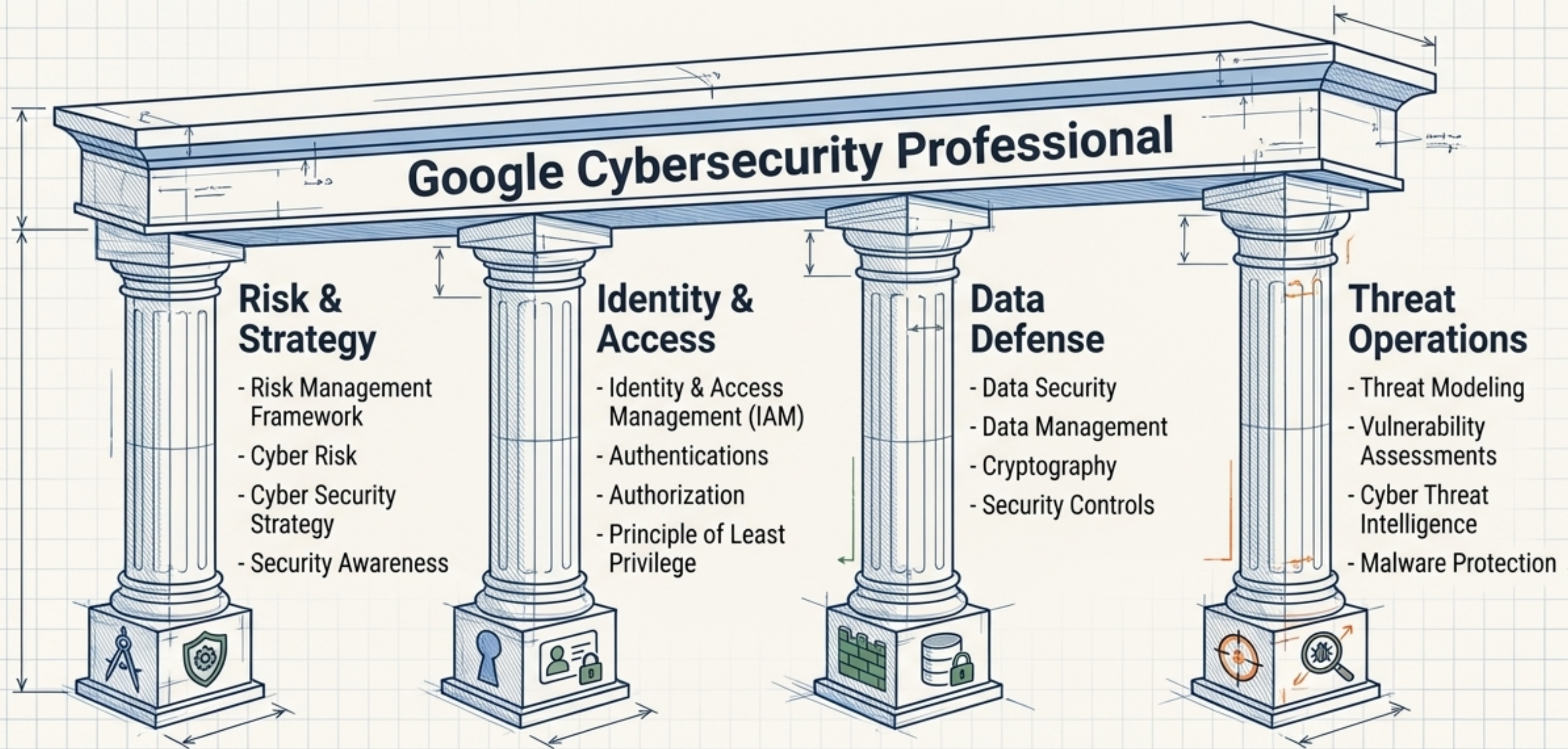
6. Attack Modeling

7. Risk & Impact Analysis

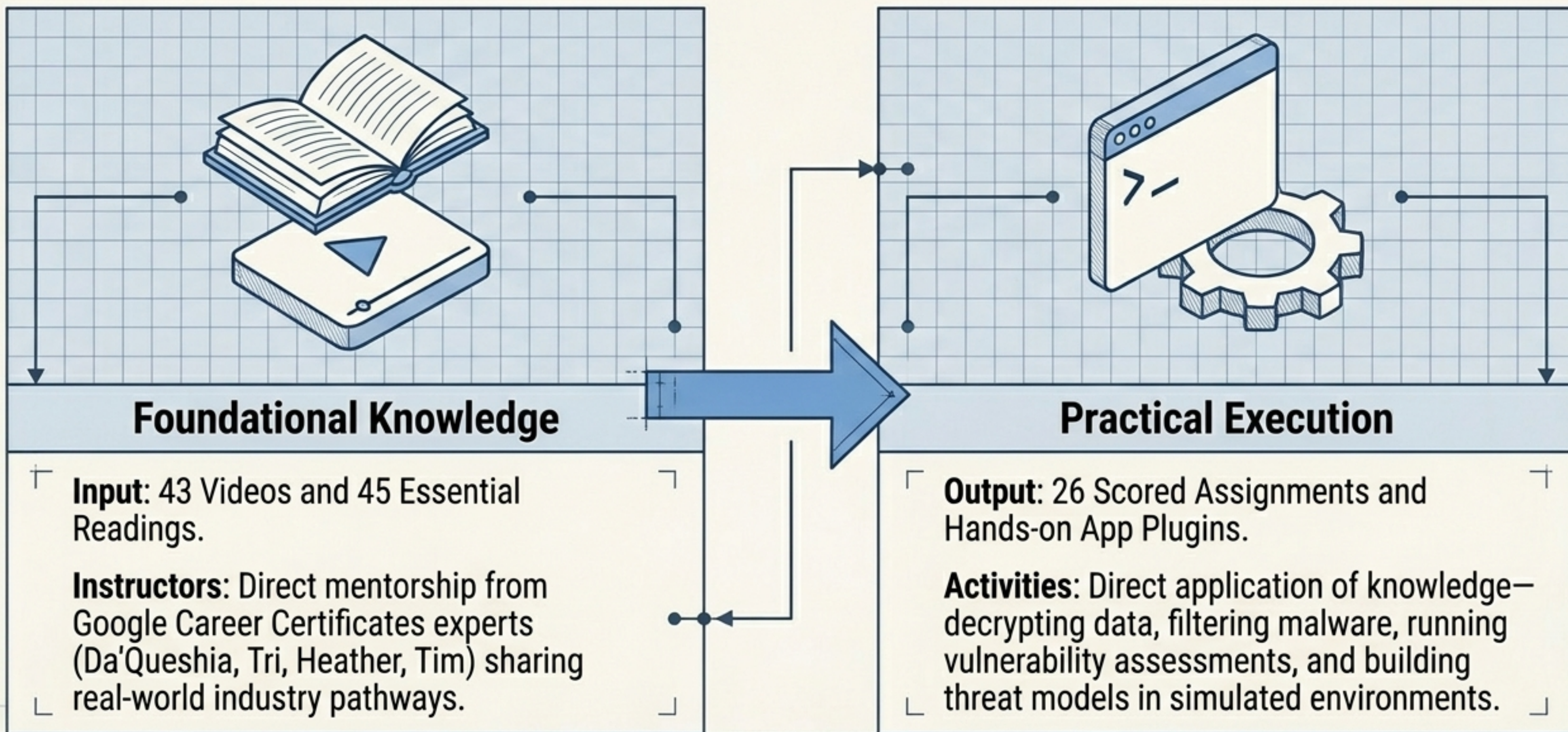
The Architect's Arsenal: Industry Frameworks Decoded

Framework	Primary Focus	Course Application
NIST CSF	Organizational Risk & Standards	Classifying assets and establishing baseline security plans (Module 1).
MITRE ATT&CK	Adversary Tactics & Techniques	Identifying how common vulnerability exposures are executed (Module 3).
OWASP	Web Application Security	Understanding the Top 10 most critical web application security risks (Module 3).
PASTA	Threat Modeling	A 7-step process for attack simulation and proactive threat analysis (Module 4).

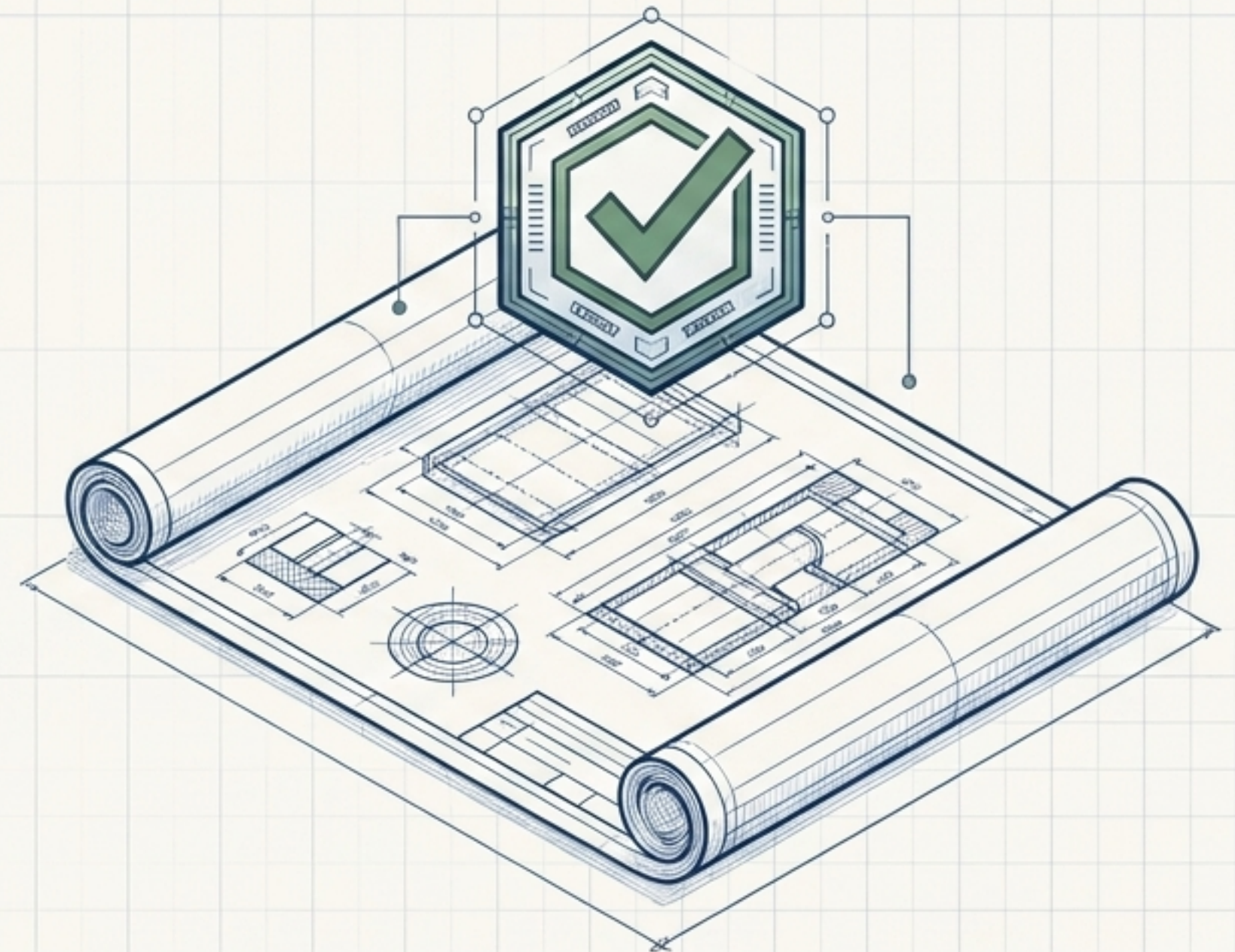
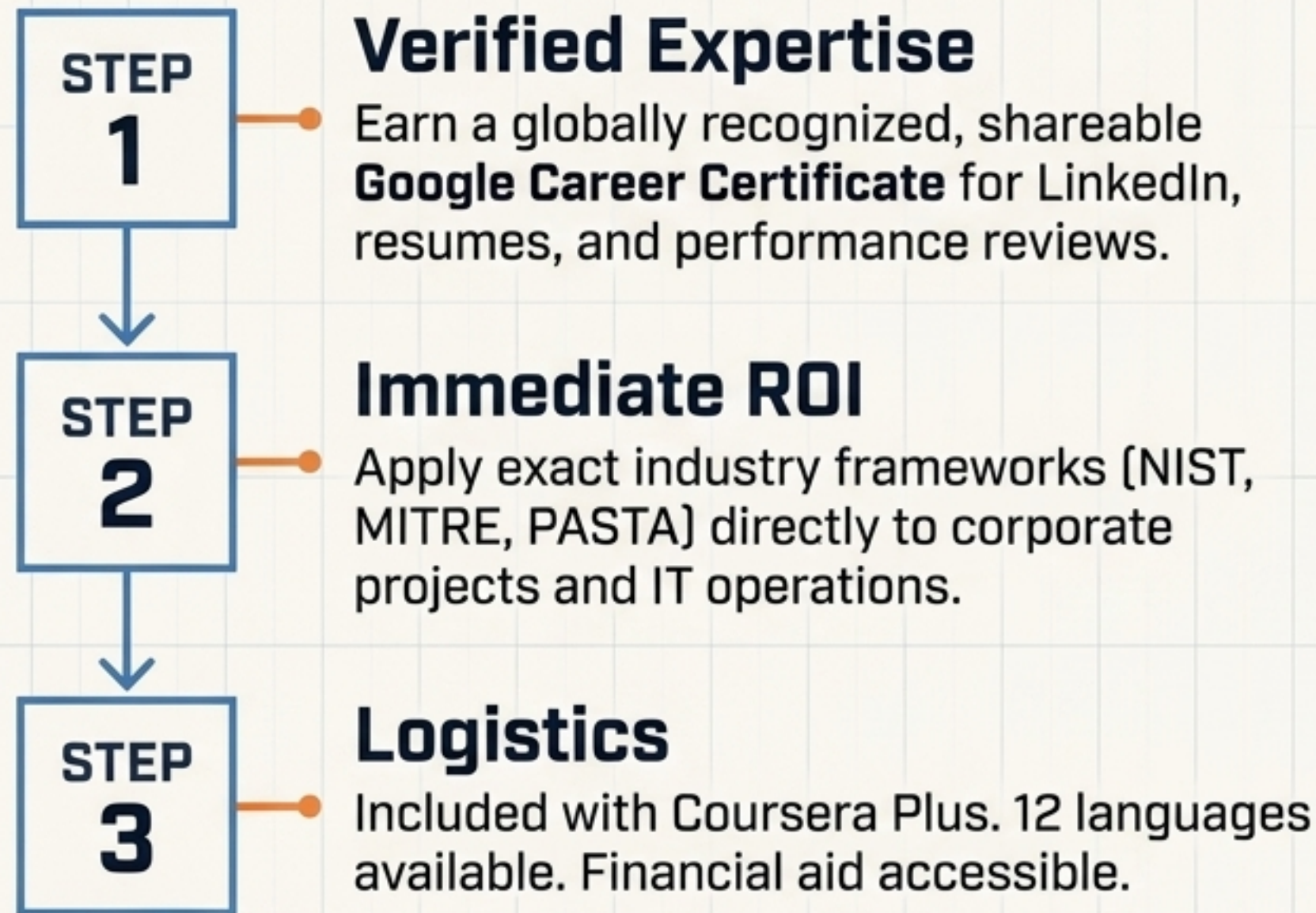
Blueprinting Your Expertise: The Skills Taxonomy



The Learner Experience: Theory Meets Execution



Mission Outcomes & Career Impact



Master the triad of Assets, Threats, and Vulnerabilities to transition from a target to a hardened digital fortress.